

Gnome Sort

35 30 4 27 2 10
↑ ↑

30 35 4 27 2 10
 ↑

30 4 35 27 2 10
 ↑

4 30 35 27 2 10
 ↑

4 30 27 35 2 10

4 30 27 35 2 10

↑

[illegible]

4 2730 2 33 10
 ↑

4 27 2 30 35 10

↑

4 2 27 30 33 10

↑

[illegible]

2 4 27 30 10 35

↑

2 4 27 10 30 35

↑

2 4 10 27 30 35

- Posso camponde com interior
- Se for menor trees e volta
- um posição

Comb Sort

↳ Subor bubble sort

↳ Problems do bubble: Números pequenos no final o algoritmo fica lento.

↳ Solução: Em vez de eu comparar os cores perto, porque não usar um gap maior.

$$(arr[i], arr[i+1]) \Rightarrow (arr[i], arr[i+gap])$$

$$g \circ p = 6$$

30 5 2 22 1 15 3

↑

87 = 4

3 5 2 2 2 1 15 30

↑ ↑ ↑ ↑ ↑

$$g^0 p = 3$$

$\begin{array}{ccccccc} & & 2 & & 3 & & \\ & \nearrow & & \nearrow & & \nearrow & \\ 1 & & 5 & & 2 & & 2 & & 2 & & 3 & & 15 & & 30 \\ \uparrow & & \uparrow & & \uparrow & & \nearrow & & & & & & & & \end{array}$

$$\text{gap} = 2$$

1 3 2 22 5 15 30

↑ ↑ ↑ ↑ ↑

$$\text{gap} = 1$$

1 3 2 15 5 22 30

↑ ↑

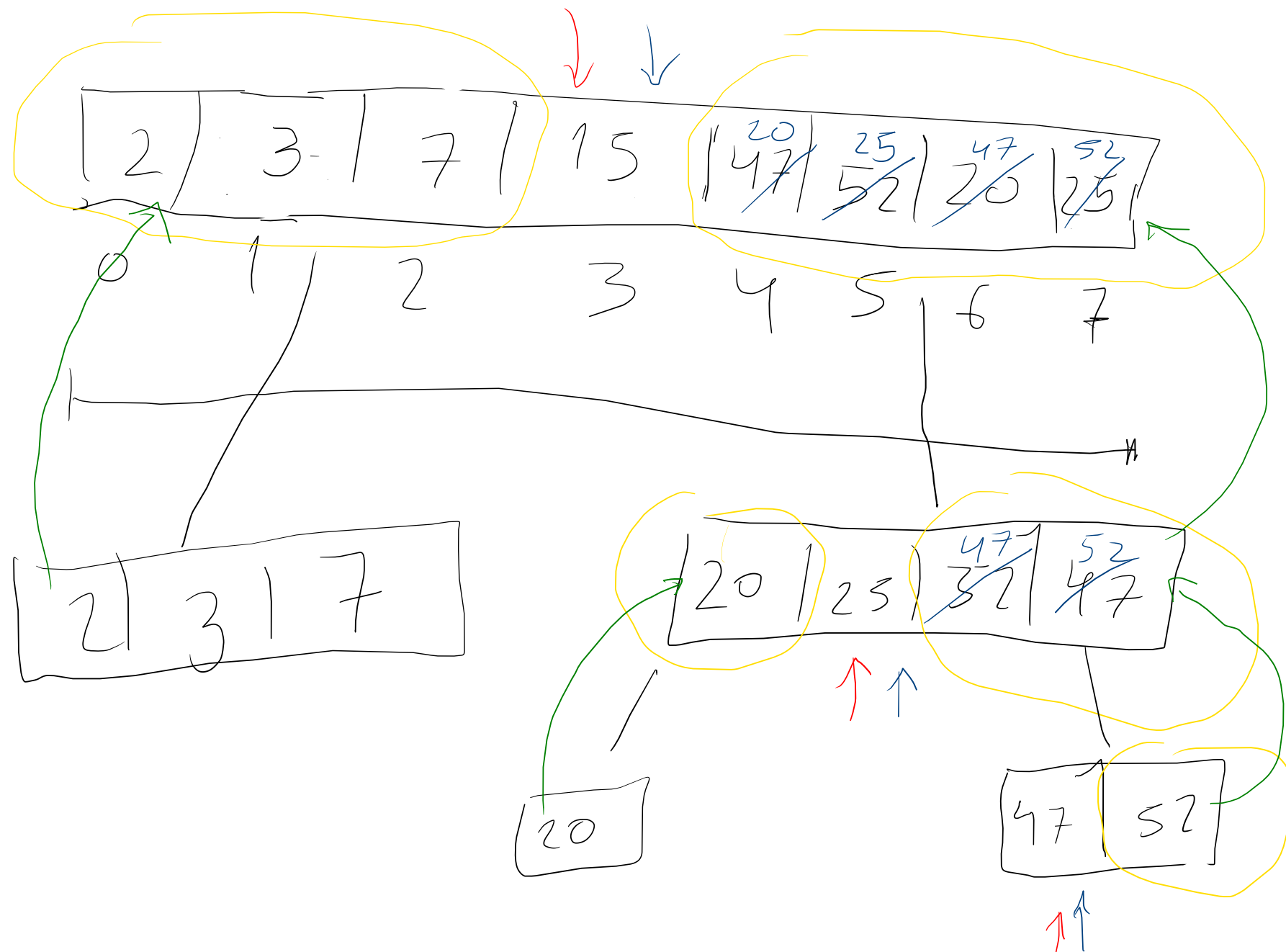
1 2 3 5 15 22 30

· Inicializo $\text{gap} = n$

· Posso em cada posição do array, comparar i e $i + \text{gap}$. Se for maior troco.

· Atualizo $\text{gap} = \text{gap} / 1.3$, até $\text{gap} = 0$.

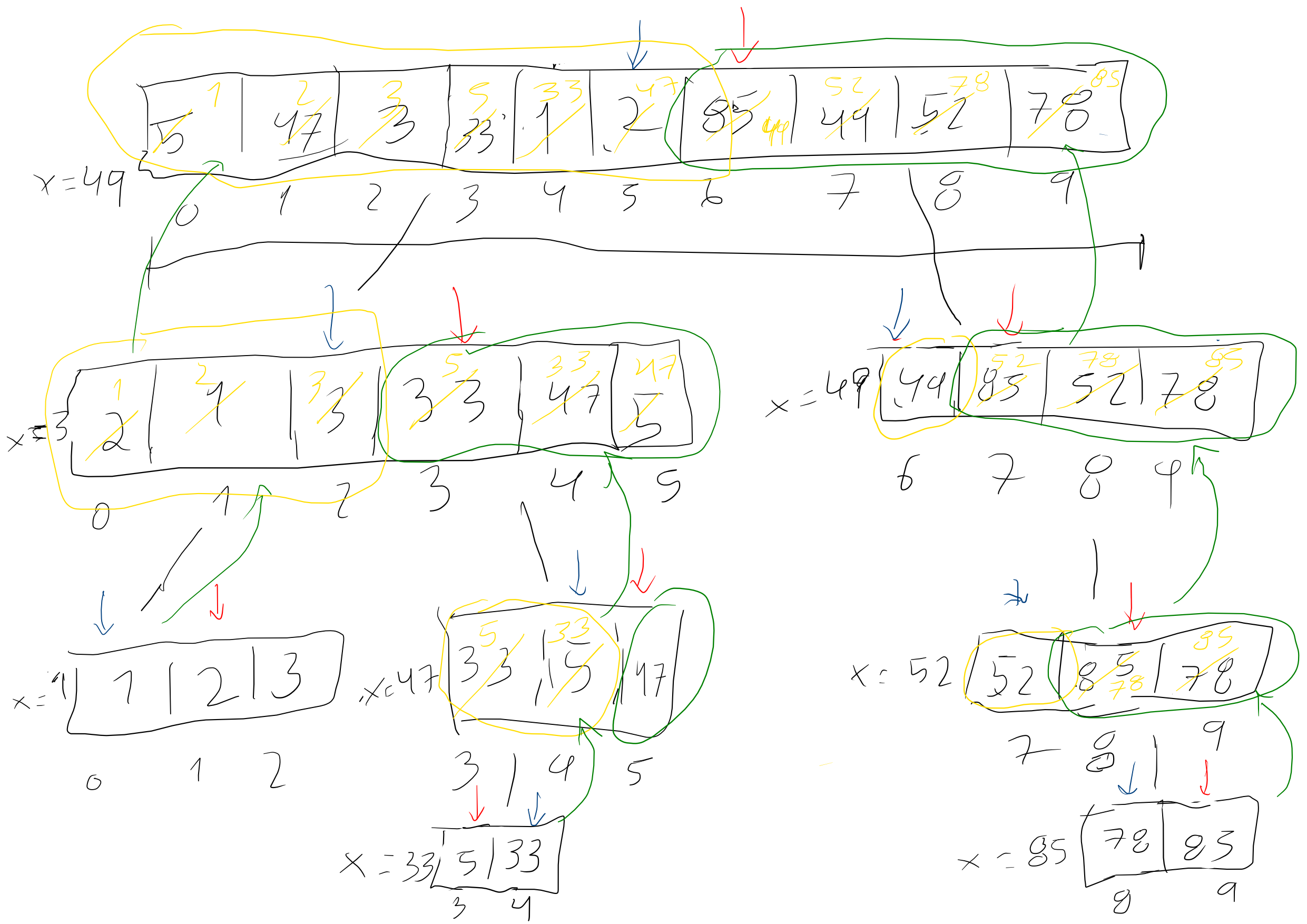
Quick Sort



· Pega o intervalo até ao menor
e acha o ponto (divide
o intervalo) e faz o mesmo
coisa para cada intervalo recursi-
vamente.

· Partição vamos passar por
os elementos comparando os
elementos do começo com o
fim, caso eles não respeitem a ordem
troque-os.

Quick Sort com Pivo



Shell Sort

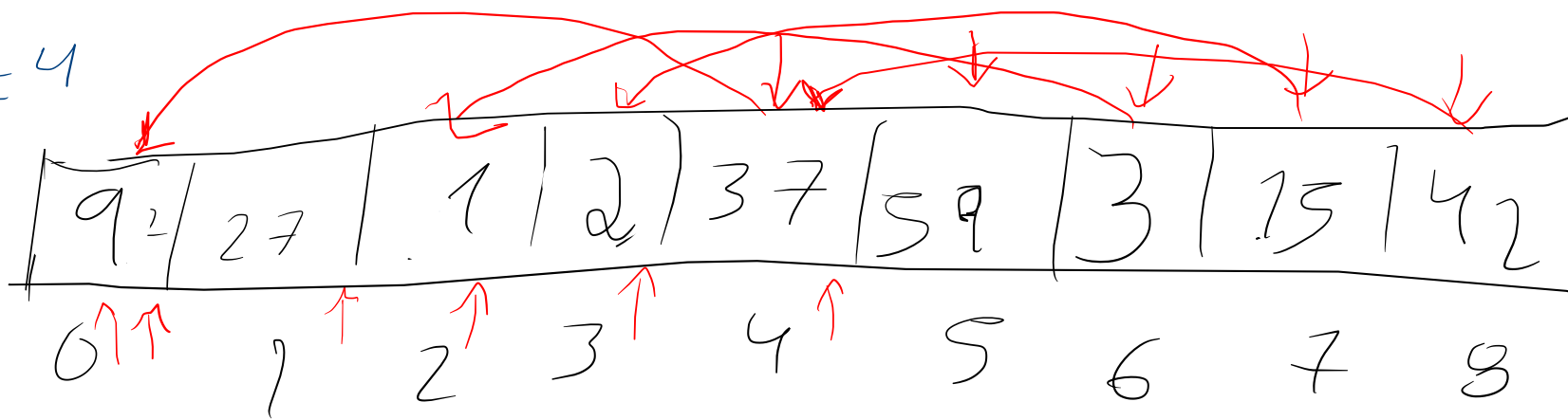
↳ Insertion Sort can gap.

↳ Knuth: $n = 3 * h + 1$

$$n = 9$$

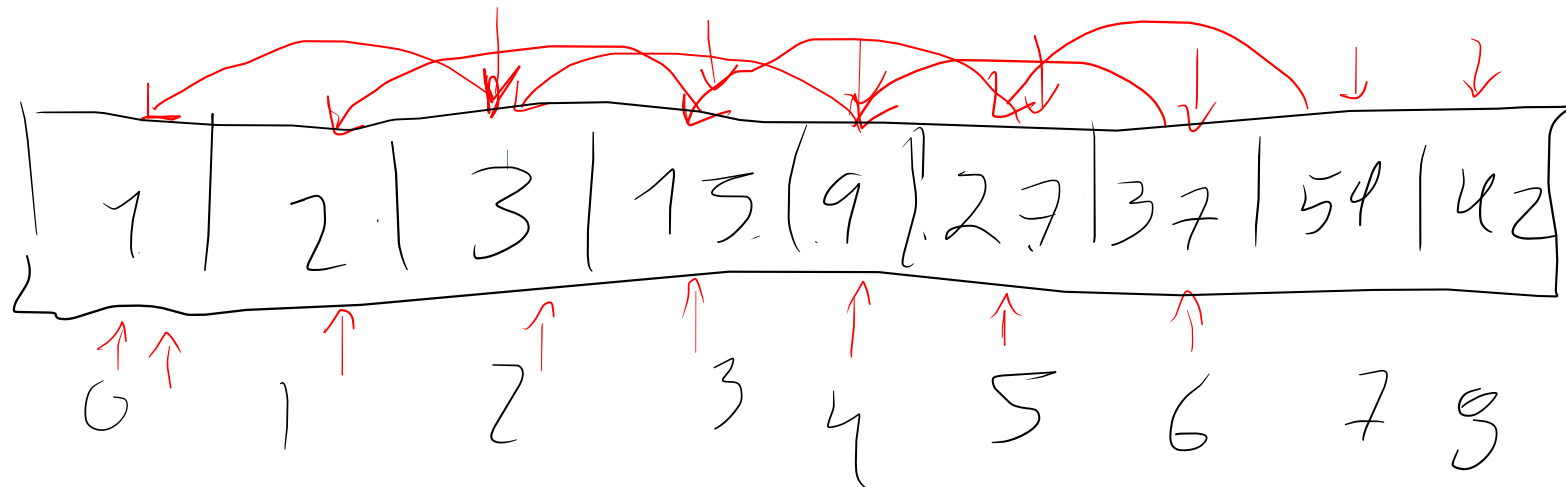
$$n = 1 \rightarrow 4 \rightarrow 13 / 3 = \textcircled{4} \rightarrow \text{gap}$$

$$h=4$$



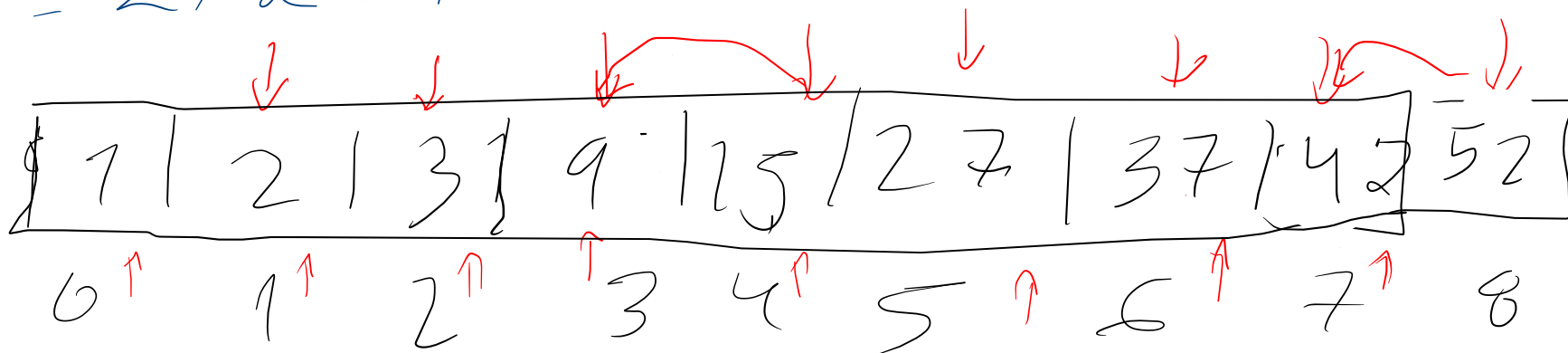
~~42~~, ~~37~~, ~~1~~
~~2~~, ~~9~~

$$h = 4 / 2 = 2$$



~~1~~, ~~2~~, ~~3~~,
~~15~~,

$$h = 2 / 2 = 1$$

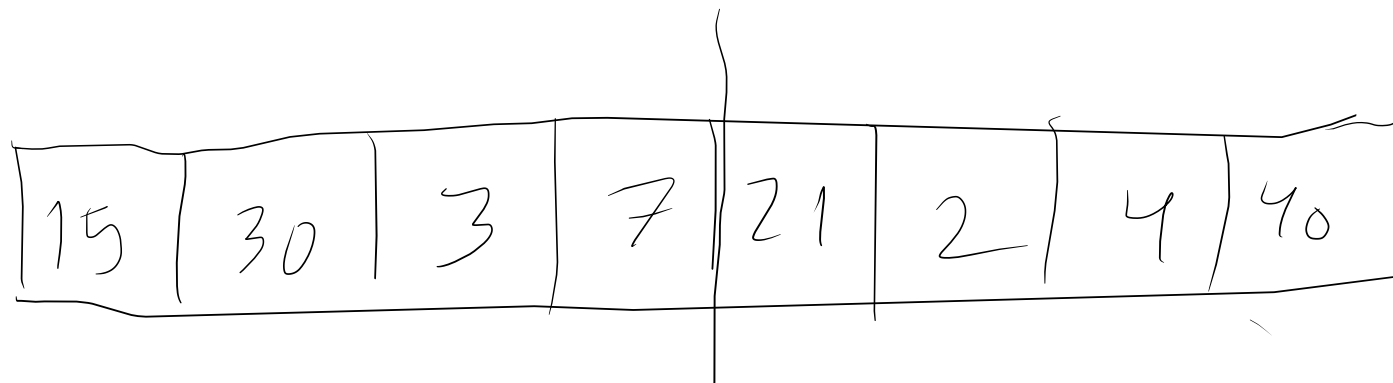


~~1~~, 42

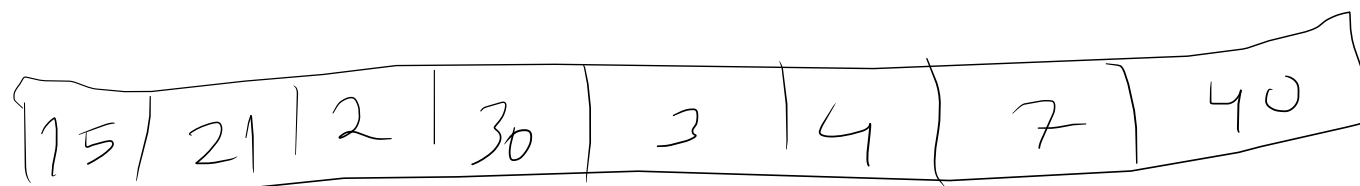
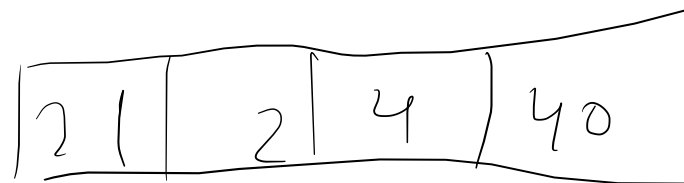
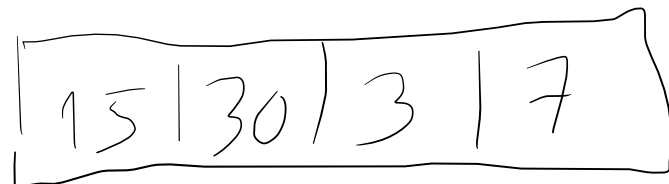
Merge Sort

↳ só funciona para potência de 2:

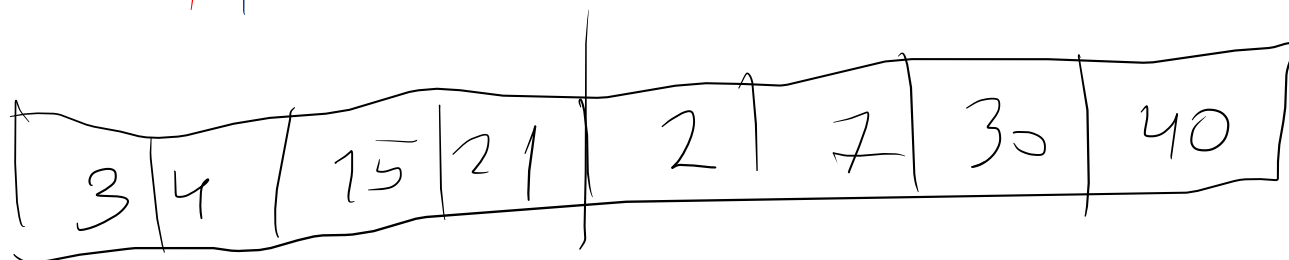
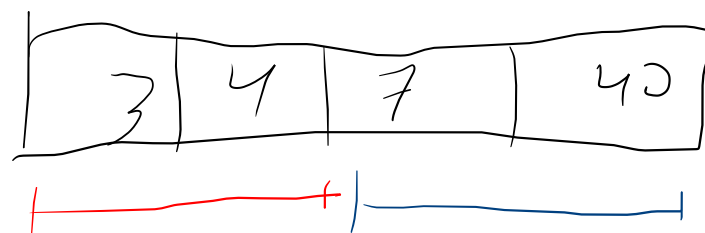
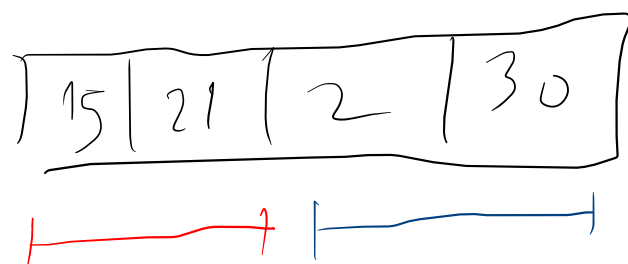
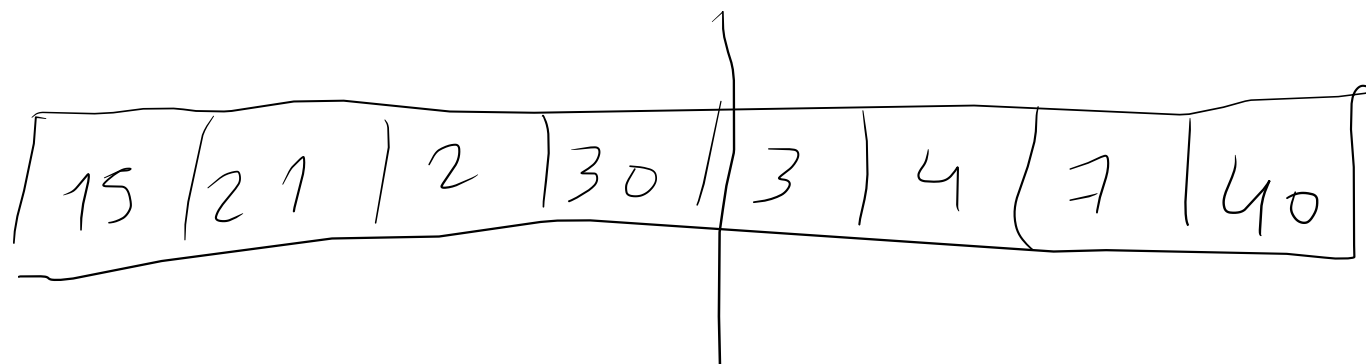
2, 4, 8, 16, ..., 2^n



• seq = 1



$$seq = 2$$



$$seq = 4$$

